

Table 1: Selections and corrections applied during NanoAOD skimming, in execution order.

Step	Operation	Kind	Condition / definition	Scope	Correction state
1	Orthogonal luminosity split	Event filter	OrthogonalEraTag == 0/1/2 (seed 12345; luminosity fractions 110/111/26)	MC 2024–2026	Before all physics corrections
2	Generator VBF phase space	Definition only	Compute GenVBFFilter for selected inclusive/VBF-filtered DY samples; current code does not call df.Filter	MC DY 2024–2026	Before corrections; no rejection
3	Base event weights	Weight definition	Luminosity, generator sign, pileup and cross-section weights; configured QCD-scale sums	MC	Pileup is applied here; no rejection
4	Golden JSON	Event filter	Accept certified run:luminosityBlock pairs from maincfg.yaml	Data	Before object corrections
5	Muon ScaRe	Object correction	Momentum scale for data/MC; MC resolution smearing; NanoAOD and beam-spot-constrained momenta	Data and MC	Before every muon selection
6	Muon FSR recovery	Object correction	FSR photon recovery implemented in corrections/muon_fsr.py	Data and MC	After ScaRe; nominal selection uses Muon_pt_FSR_corr
7	JEC/JER	Object correction	Reapply JEC; apply JER to MC; produce JER/JES Total shifts when variations are enabled	Data and MC	Before MET and jet selections
8	MET quality flags	Event filter	Logical AND of goodVertices, globalSuperTightHalo2016Filter, EcalDeadCellTriggerPrimitiveFilter, BadPFMuonFilter, BadPFMuonDzFilter, hfNoisyHitsFilter, eeBadScFilter, ecalBadCalibFilter	Data and MC	NanoAOD flags; special 2022/2023 data bad-MET redefinition
9	Single-muon trigger matching	Event filter	HLT_IsoMu24; corrected+FSR muon pT > 26 GeV; trigger-object ID 13, filter bit 8, DeltaR < 0.4	Data and MC	After muon corrections; OR over nominal/shifts
10	Good muons	Object selection	corrected+FSR pT > 15.0 GeV; eta < 2.4; medium ID; pIsoId >= 2	Data and MC	ScaRe+FSR; nominal and muon shifts
11	Exactly two muons	Event filter	Exactly two good muons; nominal and every muon shift must pass when variations are enabled	Data and MC	After ScaRe, FSR and trigger matching
12	Dimuon mass window	Event filter	50 < m_mumu < 200 GeV; nominal and every muon shift must pass	Data and MC	Selected ScaRe+FSR muons
13	Electron veto	Event filter	No electron with pT > 20 GeV, eta < 2.5 and Electron_mvIso_WP90	Data and MC	Raw NanoAOD electron kinematics
14	Muon ID/ISO/trigger SF	Weight definition	Configured correctionlib SF sources for both selected muons; central and optional up/down	MC	After muon/electron filters; no rejection
15	Jet ID and b-tag flags	Object definition	Tight jet ID; b-tag flags at eta < 2.5; b-tag event-veto flag is stored but not filtered	Data and MC	JEC/JER jets and era-specific WPs
16	Jet veto map	Object definition	Evaluate map; apply_filter=False in analysis/skim.py	Data and MC	Corrected jets; no direct event filter
17	Selected jets	Object selection	corrected pT > 20.0 GeV; eta < 4.7; tight ID; outside veto map; DeltaR(j,mu1/2) > 0.4	Data and MC	JEC/JER jets; repeated for JER/JES shifts
18	VBF candidates	Definition only	Build pair outside the horn; store HasVBF and indices	Data and MC	No VBF event filter in skim
19	Snapshot	Output	Write surviving events and nominal/shifted columns	Data and MC	Final step; categories are downstream

Table 2: Era-dependent skim settings read from the YAML configuration.

Era	Muon/jet settings	Horn veto expression	Other event-filter settings
Run3_2022	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	(abs(v_ops::eta(Jet_p4)) > 2.5 && v_ops::pt(Jet_p4) < 50.0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24
Run3_2022EE	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	(abs(v_ops::eta(Jet_p4)) >= 2.5 && v_ops::pt(Jet_p4) < 50.0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24; special bad-MET runs=362433–362760
Run3_2023	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	(abs(v_ops::eta(Jet_p4)) >= 2.5 && v_ops::pt(Jet_p4) < 50.0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24; special bad-MET runs=366727–367144
Run3_2023BPix	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	(abs(v_ops::eta(Jet_p4)) >= 2.5 && v_ops::pt(Jet_p4) < 50.0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24
Run3_2024	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	((abs(v_ops::eta(Jet_p4)) >= 2.5 && abs(v_ops::eta(Jet_p4)) < 3.0) && v_ops::pt(Jet_p4) < 50.0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24
Run3_2025	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	(abs(v_ops::eta(Jet_p4)) < 0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24
Run3_2026	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	abs(v_ops::eta(Jet_p4)) < 0	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24

Important scope note. The muons_selection, categories, and masses_regions expressions in config/Run3_*/selections.yaml are not invoked by the current analysis/skim.py. They are downstream histogram-level definitions, not NanoAOD skim event filters.